



DPUDX8RacingWheel

Provides an interface to DirectX 8.0 compatible steering wheels. The steering wheel must support force feedback. If it does not (e.g., most joysticks), **DPUDX8Joystick** must be used instead.

1. Simple Force Feedback Input:

Force Force feedback to the driver. The range is between -1 (to the left) and +1 (to the right).

2. Advanced Force Feedback Inputs:

For each of the four force-feedback subsystems *Spring*, *Damper*, *Friction*, and *Inertia* (details: see below), the following variables below *SystemName*. exist:

<i>Offset</i>	Center of axis (where no force is applied) [-1..1]
<i>DeadZone</i>	Dead zone around the offset; no force is applied within the range [Offset-DeadZone .. Offset+DeadZone] [0..1]
<i>NegativeCoefficient</i>	Linear slope of the force, if the axis value is less than <i>Offset</i> [-1..1]
<i>PositiveCoefficient</i>	Linear slope of the force, if the axis value is greater than <i>Offset</i> [-1..1]
<i>NegativeSaturation</i>	Saturation value of force, if the axis value is less than <i>Offset</i> [0..1]
<i>PositiveSaturation</i>	Saturation value of force, if the axis value is greater than <i>Offset</i> [0..1]

3. Dynamic parameters:

<i>upThreshold</i>	Threshold from which <i>y</i> values are interpreted as <i>up</i> . Default: 0.2
<i>downThreshold</i>	Threshold from which <i>y</i> values are interpreted as <i>down</i> . Default: -0.2
<i>leftThreshold</i>	Threshold from which <i>x</i> values are interpreted as <i>left</i> . Default: -0.2
<i>rightThreshold</i>	Threshold from which <i>x</i> values are interpreted as <i>right</i> . Default: 0.2

4. Static parameters:

<i>JoystickID</i>	ID of the steering wheel. 1 is the first connected steering wheel, 2 the second, etc. Default: 1
<i>NoError</i>	If set to <i>true</i> , the simulation will not be stopped if an error occurs while accessing the steering wheel. Default: <i>true</i>
<i>AlternateForces</i>	If set to <i>false</i> (the default), all subsystems are updated in every simulation step. If set to <i>true</i> , then only one of the advanced force feedback subsystems is set per simulation step. Use this setting if the device driver takes a long time to update the force feedback settings. Default: <i>true</i>



5. Outputs:

<i>x</i>	Value of the steering wheel's X axis (normally corresponds to left/right movement, between -1 (left) and +1 (right))
<i>y</i>	Value of the steering wheel's Y axis (normally corresponds to back/forward movement, between -1 (back) and +1 (forward))
<i>z</i>	Value of the steering wheel's Z axis (often corresponds to the „throttle“ control, between -1 (low) and +1 (high))
<i>rotx</i>	Value of the steering wheel's rotation around the X axis (range: -1 to 1)
<i>roty</i>	Value of the steering wheel's rotation around the Y axis (range: -1 to 1)
<i>rotz</i>	Value of the steering wheel's rotation around the Z axis (range: -1 to 1)
<i>up</i>	Boolean output, which is set to true when <i>x</i> > <i>upThreshold</i> .
<i>down</i>	Boolean output, which is set to true when <i>x</i> < <i>downThreshold</i> .
<i>left</i>	Boolean output, which is set to true when <i>y</i> < <i>leftThreshold</i> .
<i>right</i>	Boolean output, which is set to true when <i>y</i> > <i>rightThreshold</i> .
<i>Button[1...32]</i>	State of the buttons (0: unpressed, 1: pressed)
<i>Slider[1...2]</i>	Value of the sliders (range is not standardized)
<i>Pov[1...4]</i>	Position of the coolie hats („point of view hats“). Unit: 1/100 degree
<i>Pov[1...4]N</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north.
<i>Pov[1...4]NW</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north-west.
<i>Pov[1...4]W</i>	Boolean output, which is set to <i>true</i> when this coolie hat points west.
<i>Pov[1...4]SW</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south-west.
<i>Pov[1...4]S</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south.
<i>Pov[1...4]SO</i>	Boolean output, which is set to <i>true</i> when this coolie hat points south-east.
<i>Pov[1...4]O</i>	Boolean output, which is set to <i>true</i> when this coolie hat points east.
<i>Pov[1...4]NO</i>	Boolean output, which is set to <i>true</i> when this coolie hat points north-east.

6. Functional principle:

The DPUDX8RacingWheel represents an interface to any steering wheel that is supported by Microsoft DirectX 8.0. However, the steering wheel must support force feedback.

The axis range is standardized between -1 and +1. It is necessary to install Microsoft DirectX 8.0 or higher before operating the DPU.



If the connected steering wheel doesn't have so many axes, buttons, coolie hats or sliders, the corresponding DPU outputs are set to zero.

7. Force Feedback:

There are two force feedback systems which can be used with DPUDX8RacingWheel: the *simple* and the *advanced force feedback system*.

Both systems can be used at the same time, if desired.

The *simple force feedback system* consists of the input variable *Force*. The range of this variable is between -1 (maximum force towards the left) and +1 (maximum force towards the right).

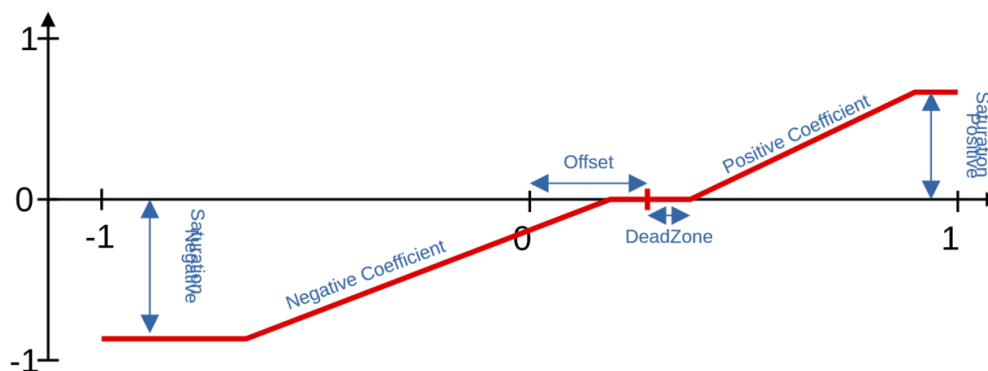
The *advanced force feedback system* consists of four modules:

- *Spring* simulates a spring which centers the steering wheel at a desired position. The force is proportional to the angular distance from the desired position.
- *Friction* makes it harder to move the wheel, independent of speed and direction.
- The *Damper* makes it harder to move the wheel, proportional to the speed of movement.
- The *Inertia* system generates can be used (using negative coefficients) to implement a “sluggish” wheel which keeps moving for a little longer even after it has been released by the driver. The force is proportional to the acceleration of the wheel.

Note: For some steering motors, high latencies have been observed when using the advanced force feedback system. Please verify the *SysProfileTriggerMS* time of your instance of DPUDX8RacingWheel after configuring the advanced force feedback system.

For each system, a piecewise linear equation is defined using six input parameters. It maps the input value (position, speed or acceleration, normalized to the range -1..1) to an output force (also in normalized range -1..1).

The equation is defined as follows:





The meaning of each parameter is in detail:

<i>Offset</i>	Center of axis (where no force is applied). Example: For the <i>Spring</i> , it's the position to which the wheel centers.
<i>DeadZone</i>	Dead zone around the offset; no force is applied within the range $[Offset-DeadZone .. Offset+DeadZone]$ Example: For the <i>Spring</i> , this is an area around the <i>Offset</i> position where the wheel can be moved without any resistance.
<i>NegativeCoefficient</i>	Linear slope of the force, if the axis value is less than <i>Offset</i> . Can be in range -1..1, where negative values reverse the direction of the effect. Example: For the <i>Spring</i> , this is the strength of the spring while moving to the left of the <i>Offset</i> position.
<i>PositiveCoefficient</i>	Linear slope of the force, if the axis value is greater than <i>Offset</i> . Can be in range -1..1, where negative values reverse the direction of the effect. Example: For the <i>Spring</i> , this is the strength of the spring while moving to the right of the <i>Offset</i> position.
<i>NegativeSaturation</i>	Saturation value of force, if the axis value is less than <i>Offset</i> . It is always given as a positive value (0..1). Example: For the <i>Spring</i> , this is the maximum strength of the spring on the left side of the <i>Offset</i> position.
<i>PositiveSaturation</i>	Saturation value of force, if the axis value is greater than <i>Offset</i> . It is always given as a positive value (0..1). Example: For the <i>Spring</i> , this is the maximum strength of the spring on the right side of the <i>Offset</i> position.

8. Example:

Integration of a Logitech racing wheel (G29) with gear shifter:

```
DPUDX8RacingWheel ~Joy
{
    TreeviewGroup = "Mockup";
    Computer = {MOCKUP};
    Index = 10;
};

DPUJoystickConsole ~Mockup
{
    TreeviewGroup = "Mockup";
    Computer = {MOCKUP};
    Index = 12;
    JoystickAMin = -1;
    JoystickAMax = 1;
    JoystickBMin = 1;
    JoystickBMax = -1;
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
JoystickCMin = 1000.0;
JoystickCMax = -1000.0;
HShift = true; # Für H-Schaltung
SteeringLinGain1 = 7.85;
SteeringLinGain2 = 7.85;
BrakePedalGain = 5;

};
Connections =
{
    ~Joy.x          -> ~Mockup.JoystickX_In,
    ~Joy.y          -> ~Mockup.JoystickA_In,
    ~Joy.rotz       -> ~Mockup.JoystickB_In,
    ~Joy.Slider1    -> ~Mockup.JoystickC_In,
    ~Joy.Button5    -> ~Mockup.GearDown_In,
    ~Joy.Button6    -> ~Mockup.GearUp_In,
    ~Joy.Button13   -> ~Mockup.ButtonGear1,
    ~Joy.Button14   -> ~Mockup.ButtonGear2,
    ~Joy.Button15   -> ~Mockup.ButtonGear3,
    ~Joy.Button16   -> ~Mockup.ButtonGear4,
    ~Joy.Button17   -> ~Mockup.ButtonGear5,
    ~Joy.Button18   -> ~Mockup.ButtonGear6,
    ~Joy.Button19   -> ~Mockup.ButtonGearR,

    ~Mockup.AcceleratorPedal -> Mockup.AcceleratorPedal,
    ~Mockup.BrakePedal      -> Mockup.BrakePedal,
    ~Mockup.ClutchPedal     -> Mockup.ClutchPedal,
    ~Mockup.SteeringWheel   -> Mockup.SteeringWheel,
    ~Mockup.Gear            -> Mockup.Gear
};
```